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Level Design

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1. Introduction

The Level Design system made is to provide gameplay environments that support the core mechanics, pacing, and cooperative gameplay experience of Parallax. The levels are designed to combine movement-based traversal, puzzle solving, and asymmetric cooperation into a cohesive gameplay experience.

The level design defines how players navigate the environment, interact with gameplay elements, and utilize their character-specific abilities in order to progress through the game.

The level design follows a modular structure, where reusable gameplay components such as buttons, keys, movable objects, activation systems, and traversal obstacles can be combined in different ways across levels. This helps improve maintainability, scalability, and flexibility during development while allowing gameplay mechanics to be introduced progressively throughout the game.

1.1. Requirements

Requirements for the Level Design system, as outlined in **Appendix C**, define the gameplay structure and progression goals needed to support the intended cooperative gameplay experience.

Functional Requirements

- 1) Cooperative Puzzle Solving (**E2-US1**)
- 2) Role-Specific Gameplay Mechanics (**E2-US2**)
- 3) Environmental Interaction (**E2-US3**)
- 4) Progressive Gameplay Introduction (**E2-US4**)
- 5) Traversal and Movement Challenges (**E2-US5**)

The functional requirements focus on creating gameplay scenarios that encourage communication, cooperation, and coordinated problem solving between players. These requirements ensure that both player roles contribute unique abilities necessary for progression throughout the game.

Non-Functional Requirements

- 1) Levels should provide intuitive environmental guidance (**Usability - Must**)
- 2) Levels should encourage replayability and experimentation (**Gameplay - Should**)
- 3) Puzzle mechanics should support gradual learning progression (**Usability - Must**)

The non-functional requirements focus on usability, pacing, and player experience. The level design was therefore structured to gradually introduce gameplay mechanics while maintaining clear progression, readable environments, and opportunities for experimentation and cooperation.

2. Design

The level design of Parallax was created to support cooperative gameplay, asymmetric player interaction, and gradual gameplay progression throughout the player experience. The environments, puzzles, and traversal sections were designed to work together as part of a structured learning progression where new mechanics are introduced over time and later combined into more advanced challenges.

A major focus of the level design was to encourage communication and cooperation between the two player roles while maintaining varied gameplay scenarios and recognizable environments. Each level was therefore designed around specific gameplay goals, introducing new mechanics while reinforcing previously learned interactions and traversal systems.

The following sections describe the core design principles used throughout the development of the game levels.

2.1. Gameplay Flow and Progression

The gameplay flow of **Parallax** was designed around gradual learning progression, where players are introduced to new mechanics over time instead of being required to understand all gameplay systems at once. Each level introduces new gameplay elements while simultaneously reinforcing mechanics learned in previous levels. This structure allows players to naturally build familiarity with movement, interaction, and cooperative gameplay throughout the game.

The intended progression assumes that players complete the levels in their designed order. However, during the current stage of development, players are not restricted from accessing later levels directly. As a result, players may encounter mechanics or puzzle combinations before being introduced to the foundational gameplay systems intended to teach them.

The progression system combines puzzle-oriented gameplay with traversal-based challenges that require players to utilize both movement and communication in order to advance. Early levels focus on introducing core mechanics and basic interactions in controlled scenarios, while later levels increase complexity by combining multiple mechanics into larger cooperative challenges.

Difficulty progression was designed around mechanic combination rather than simply increasing obstacle density or player punishment. As players progress through the game, levels require improved coordination, timing, traversal precision, and understanding of the asymmetric gameplay systems. Previously introduced mechanics are reused in more advanced situations, encouraging players to apply existing knowledge in new ways.

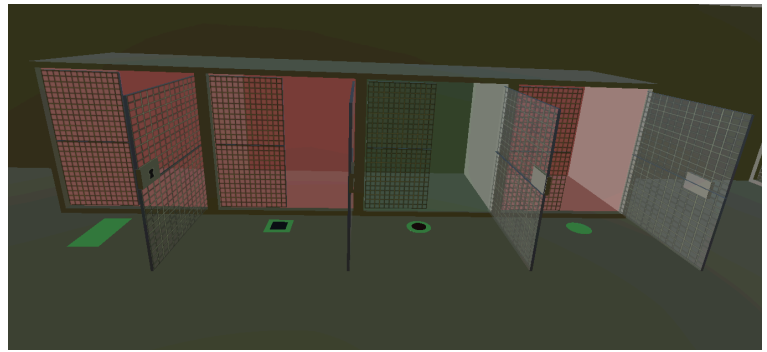


Figure 1: Example of asymmetric cooperation where the Human player identifies the correct route while the Cat player navigates the puzzle environment.

The levels were also structured to maintain pacing variation throughout the gameplay experience. Puzzle sections, traversal challenges, and cooperative interactions are alternated in order to create gameplay variety while still reinforcing the core gameplay loop centered around cooperation and environmental interaction.

2.2. Cooperative Design

The cooperative gameplay of **Parallax** is centered around asymmetric player roles, where both the Human and Cat player possess unique abilities required for progression throughout the levels. Rather than allowing both players to solve challenges in the same way, the level design was intentionally structured to create situations where players depend on one another in order to continue.

Different gameplay scenarios are designed around the strengths and limitations of each role. The Human player is primarily responsible for interacting with larger environmental objects such as buttons, movable boxes, and activation mechanisms, while the Cat player is able to access elevated areas, narrow spaces, and hard-to-reach locations within the environment.

This asymmetric structure creates gameplay situations where one player may have information, access, or interaction capabilities unavailable to the other player. As a result, communication and coordination become a major parts of the gameplay experience, requiring players to share information and cooperate in order to solve puzzles and overcome challenges.

2.3. Puzzle Design

The puzzle design of **Parallax** was created to support cooperative gameplay, environmental interaction, and gradual mechanic progression throughout the game. Puzzle sections are integrated directly into the level environments, requiring players to utilize movement, communication, and role-specific abilities in order to progress.

Rather than relying on isolated puzzle rooms, challenges are designed as part of the surrounding environment. Players are required to interact with gameplay elements such as keys, buttons, movable objects, moving platforms, and traversal routes in order to unlock progression or create new pathways through the levels.

Many puzzles are built around asymmetric gameplay situations where the two player roles must contribute differently to the solution. Certain interactions may only be accessible to the Human player, while other areas or objectives are designed specifically for the Cat player. This creates puzzle scenarios where communication and coordination become necessary in order to successfully complete objectives.

The puzzle structure also encourages experimentation and replayability by allowing players to approach some challenges in different ways depending on their understanding of the mechanics and their level of coordination. Earlier levels introduce puzzle mechanics in relatively controlled scenarios, while later levels combine multiple mechanics into larger and more demanding cooperative challenges.

The overall puzzle design therefore focuses on:

- environmental interaction
- asymmetric cooperation
- traversal-based challenges
- communication between players
- experimentation and replayability

2.4. Environment Design

The environments in **Parallax** were designed around recognizable real-world locations in order to create intuitive and readable gameplay spaces for the player. Familiar environments such as kitchens, playgrounds, construction sites, and industrial structures allow players to naturally understand the purpose of environmental objects and gameplay elements without requiring extensive tutorials or direct instruction.

Environmental design was used as a method of communicating gameplay functionality through visual cues and object placement. Interactive elements such as buttons, doors, keys, lights, platforms, and traversal routes were positioned in ways that encourage players to investigate and experiment with the environment. This allows players to learn mechanics through gameplay interaction rather than through explicit instruction.

The environments were also structured to support navigation and gameplay flow throughout the levels. Lighting, pathways, object placement, vertical landmarks, and environmental composition were used to subtly guide player attention toward progression areas while still allowing room for exploration and experimentation.

Asymmetric gameplay was further reinforced through the environmental design by creating areas that are experienced differently depending on the player role. Certain traversal paths, viewpoints, or interaction opportunities are designed specifically for either the Human or Cat player, encouraging cooperation and communication between players in order to progress.

In addition to supporting gameplay functionality, the environments contribute to the overall atmosphere and pacing of the game by combining traversal, puzzle solving, and exploration within visually distinct gameplay spaces.

2.5. Level Structure

The level structure of **Parallax** follows a progression designed to guide players through increasingly complex gameplay scenarios while still allowing freedom for exploration and experimentation. Each level introduces new mechanics or gameplay situations that build upon systems and interactions learned in earlier sections of the game.

The progression between levels was intentionally designed around gradual escalation of mechanic complexity. Early levels focus on introducing core gameplay concepts such as asymmetric interaction, environmental puzzles, and traversal mechanics, while later levels combine multiple systems into larger cooperative challenges requiring greater coordination and communication between players.

To support both replayability and efficient development, many gameplay interactions were implemented using reusable modular systems. Gameplay elements such as buttons, moving platforms, doors, keys, movable objects, and activation systems were designed as reusable components that could be combined and adjusted across multiple levels. This allowed new puzzle scenarios and traversal challenges to be created without requiring entirely new gameplay systems for each level.

The overall level structure was designed to balance:

- gameplay progression
- cooperative interaction
- traversal challenges
- environmental exploration
- puzzle complexity

This structure allows the gameplay experience to gradually evolve throughout the game while maintaining consistent core mechanics and cooperative gameplay principles.

2.5.1. Level one: Kitchen

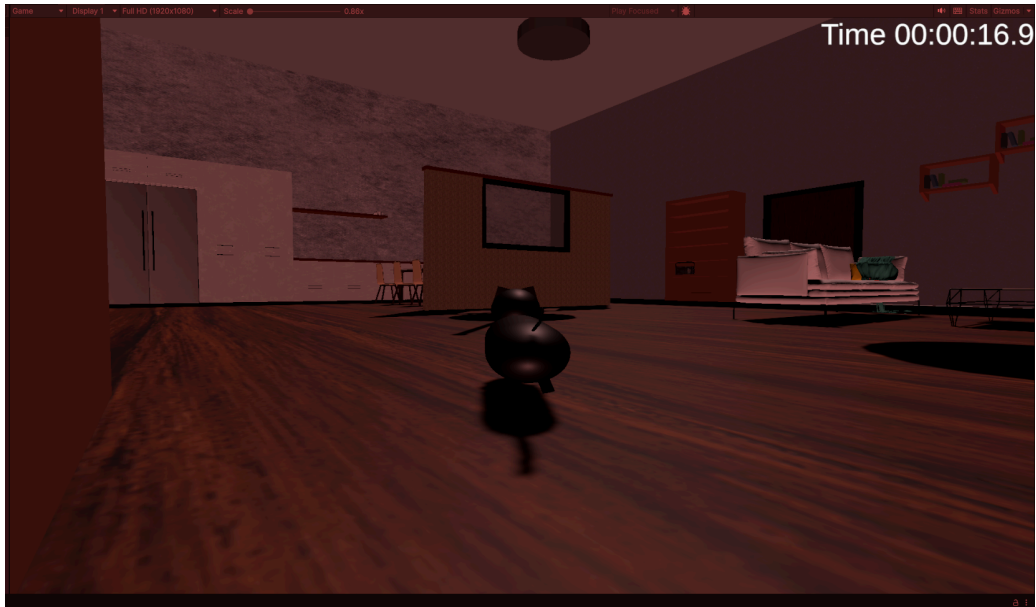


Figure 2: Early gameplay screenshot from the Kitchen level showing the Cat player's perspective in the darkened environment before power is restored.

The first level of the game, called **Kitchen**, serves as an introduction to the game world and the core gameplay structure of Parallax. The level takes place inside a house kitchen environment designed to establish the atmosphere and visual style used throughout the game. Furniture, household objects, and environmental props from Unity asset store [1] are placed in recognizable locations to create a familiar and readable environment for the player.

Although both players begin in the same area, the environment is experienced differently depending on the selected role. The level starts in near darkness, heavily limiting the Human player's visibility, while the Cat player is still able to navigate the environment through its enhanced night vision ability.

This difference in perception introduces the asymmetric relationship between the two roles early in the game. While exploring the environment, the Cat player can access elevated shelves containing a key that the Human player cannot reach. The Cat is also able to locate the fuse box required to restore power to the kitchen.

Progression through the level requires both players to contribute differently to the solution. The Cat retrieves the key and delivers it to the Human player, allowing the fuse box to be unlocked and activated. Once the power has been restored, the environment becomes fully visible and progression through the level can continue.

The level was intentionally designed as a controlled introduction to the core gameplay systems before later levels introduce more demanding traversal sections and cooperative puzzle combinations.

2.5.2. Level two: Playground

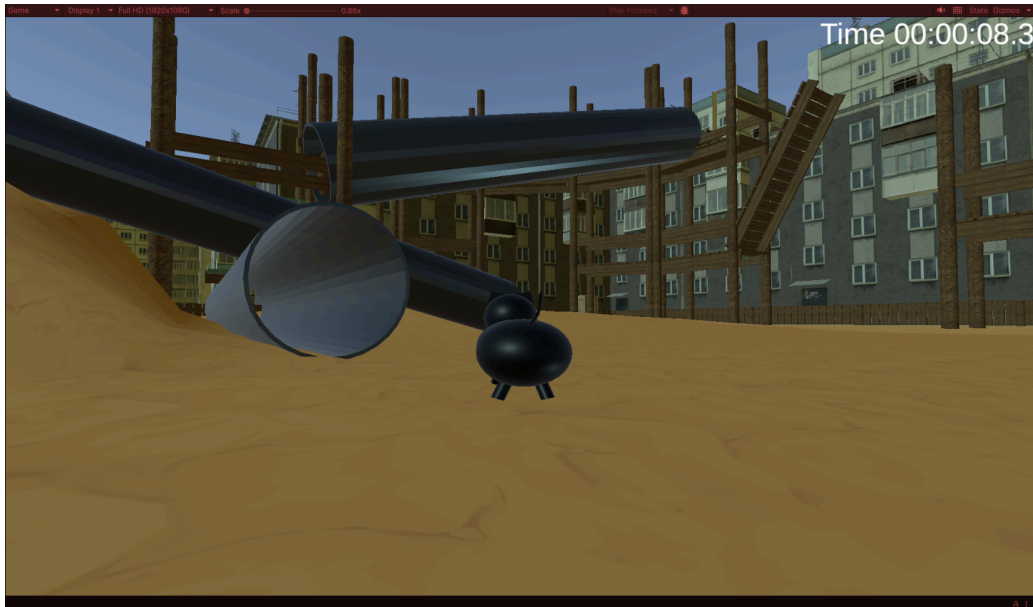


Figure 3: Gameplay screenshot from the playground level showing the vertical tower structure and traversal-focused environment.

The second level of Parallax takes place in a playground environment centered around a large climbable tower structure. While the Kitchen level focuses primarily on introducing the asymmetric player roles, this level expands the gameplay by introducing traversal-oriented challenges and more advanced environmental interaction.

The primary objective of the level is for both players to reach the top of the tower and access the finish zone. Progression requires players to move vertically through the environment using climbing, jumping, and coordinated interaction with gameplay objects.

Movable boxes are introduced as a new gameplay mechanic in this level. Players must reposition objects within the environment to create pathways, reach elevated areas, and assist one another in navigating the tower structure. This expands the interaction system beyond simple activation mechanics and introduces more dynamic puzzle-solving scenarios.

The level also introduces moving platforms that require players to activate environmental mechanisms and coordinate their movement timing in order to progress. In several sections, one player must create or activate traversal opportunities for the other player before being able to continue themselves.

Verticality plays a central role throughout the playground environment. The tower structure was designed to guide players upward through increasingly complex traversal sections while gradually combining mechanics introduced earlier in the game. As players ascend, the gameplay places greater emphasis on positioning, timing, and understanding of the cooperative systems.

Compared to the Kitchen level, the playground environment shifts the gameplay focus toward traversal and environmental navigation while continuing the gradual progression of gameplay complexity established earlier in the game.

2.5.3. Level three: Construction Site



Figure 4: Gameplay screenshot from the construction site level showing the traversal-focused environment and cooperative platform mechanics.

The third level of Parallax takes place in a construction site environment and combines gameplay mechanics introduced in earlier levels into larger and more demanding cooperative challenges. At this stage, players are expected to understand the core gameplay systems, allowing the level design to place greater emphasis on timing, positioning, and multi-step puzzle sequences.

The level combines traversal, object interaction, moving platforms, and role-specific responsibilities within the same gameplay scenarios. Earlier mechanics are no longer introduced individually, but instead reused together in more complex situations requiring players to coordinate actions across different parts of the environment.

A central mechanic introduced in this level is a movable platform that requires direct cooperation between both players. One player must stand on the platform while the other activates a control mechanism to move it. This creates sections where players become temporarily separated and must rely on communication and timing in order to continue progression.

Responsibilities between the two player roles change throughout the level depending on the puzzle structure. In some situations, the Human player operates mechanisms while the Cat navigates traversal sections, while other scenarios reverse these responsibilities to create variation in gameplay flow.

The construction site environment was intentionally designed around vertical traversal and movement challenges. Scaffolding, suspended platforms, climbable structures, and construction

materials are used to create layered traversal paths and environmental obstacles that support the increasing gameplay complexity.

Compared to previous levels, puzzle sequences in the construction site require players to consider positioning, traversal timing, and interaction order simultaneously. The level therefore acts as a transition toward larger and more mechanically demanding gameplay scenarios while continuing the cooperative gameplay progression established throughout the earlier levels.

2.5.4. Level four: The Tower

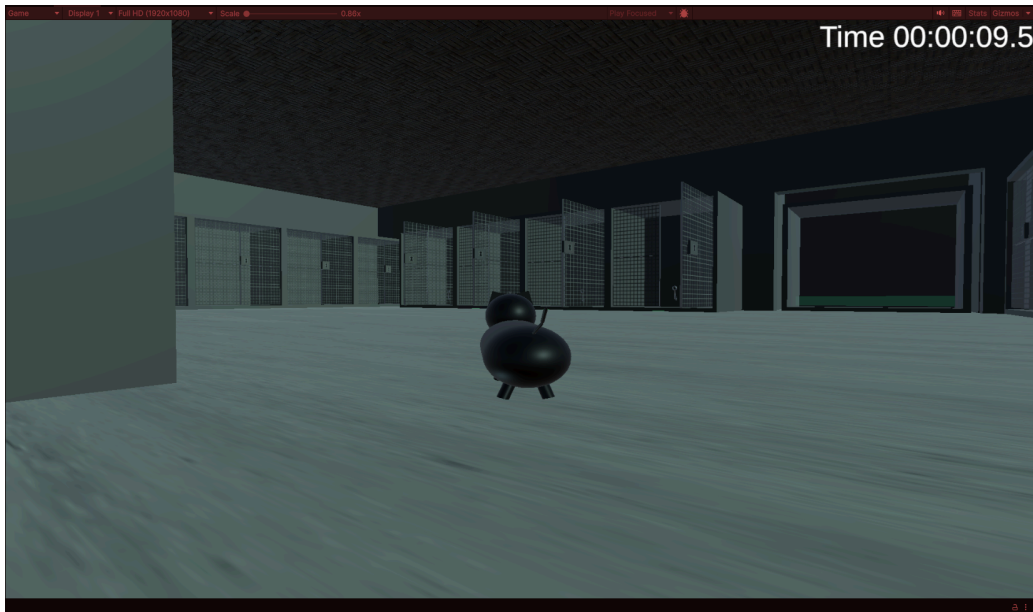


Figure 5: Gameplay screenshot from the Tower level showing the basement puzzle area and the large-scale vertical environment.

The fourth level of Parallax, called **Tower**, functions as the culmination of the gameplay mechanics introduced throughout the earlier levels. It is designed as the largest and most mechanically demanding level in the game, combining traversal, environmental interaction, timing-based puzzles, and asymmetric gameplay into a multi-stage cooperative experience.

The level takes place inside a multi-story tower structure where players must progress upward through a sequence of increasingly difficult puzzle floors. Each floor introduces either a new gameplay scenario or a more advanced variation of previously encountered mechanics, requiring players to apply knowledge gained throughout the earlier parts of the game.

Traversal and environmental interaction play a major role throughout the level. Moving platforms, vertical climbing sections, box movement, traversal routes, and role-specific interactions are combined into larger puzzle sequences that require players to coordinate multiple actions in the correct order.

The tower structure was intentionally designed to create a strong sense of progression as players ascend through the environment. Each floor acts as a contained gameplay challenge with its own

traversal layout, puzzle structure, and environmental composition. This allows gameplay difficulty to scale gradually while maintaining a consistent progression flow throughout the level.

As the temporary final level of the game, Tower requires players to combine previously learned mechanics and apply them under more demanding gameplay conditions. The level therefore functions as both the mechanical and gameplay climax of Parallax by testing the player's understanding of traversal, interaction systems, timing, and cooperative problem solving.

2.5.5. Implementation

The levels in **Parallax** were implemented in Unity using a combination of scene-based level construction, modular gameplay prefabs, and reusable interaction systems. Each level was built as a separate gameplay environment containing player spawn points, environmental assets, colliders, lighting, puzzle objects, and progression triggers.

A major focus of the implementation was reusability. Gameplay elements such as buttons, doors, keys, movable boxes, moving platforms, activation systems, and kill zones were implemented as modular components that could be reused and combined across multiple levels. This allowed puzzles to be created by connecting existing gameplay objects rather than implementing unique systems for each level.

Puzzle progression was implemented through combinations of interactable objects and activation logic. For example, the Kitchen level uses a key and fusebox interaction to control the lighting state of the environment, while later levels combine buttons, moving platforms, movable boxes, and traversal paths to create more complex cooperative challenges.

Traversal sections were implemented through level geometry, object placement, moving platforms, and vertical structures. These elements were arranged to guide players through the intended route while still allowing experimentation with movement and puzzle solutions.

Kill zones were added to levels with vertical traversal in order to handle fail states. When a player falls from the intended play area or enters a restricted zone, the kill zone resets the player or level state, preventing soft-locks and allowing the challenge to be attempted again.

```

protected override void OnPlayerEnter(GameObject
player)
{
    Debug.Log(_contactMessage);

    if (player.TryGetComponent(out RoleController rc))
    {
        Role flag = rc.CurrentRole switch
        {
            CharacterRole.Cat => Role.Cat,
            CharacterRole.Human => Role.Human,
            _ => throw new Exception("Invalid role")
        };

        if ((_activeRoles & flag) != 0)
        {
            GetComponent<Collider>().enabled = false;
            LevelManager.Instance.RestartLevel();
        }
    }
}

```

Code Snippet 1: Example implementation of a kill zone used to restart the level when a player enters a restricted area.

Lighting and visibility were also used as part of the implementation. In the Kitchen level, the environment begins in darkness to support the asymmetric gameplay between the Human and Cat roles. After the fusebox has been activated, the lighting state changes, revealing the environment more clearly and allowing both players to continue progression.

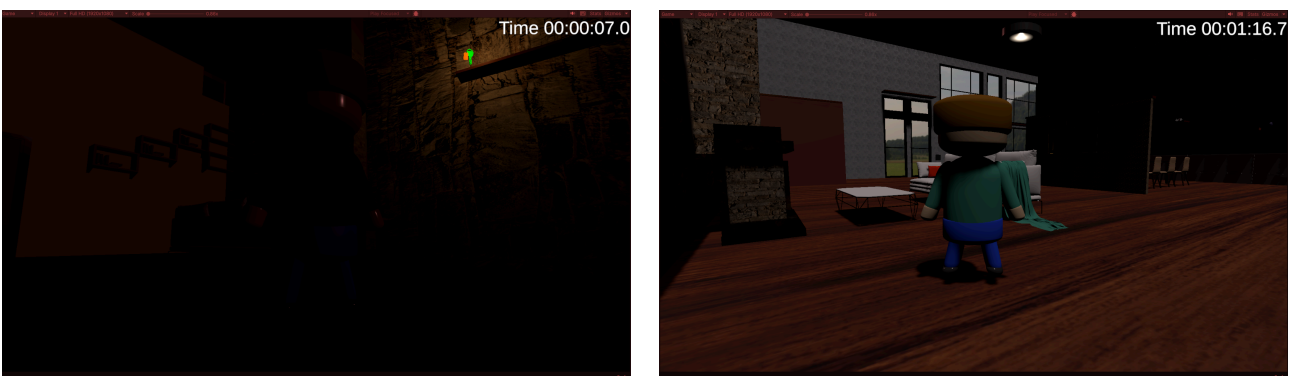


Figure 6: Example of asymmetric lighting in the first level, where the cat player assists the human player with navigation and visibility.

3. Testing Execution

Testing of the level design was primarily done through playtesting throughout development. Since many aspects of gameplay experience, level flow, and cooperative interaction are difficult to measure directly, manual testing was necessary in order to evaluate how players experienced the levels during gameplay.

Testing focused on whether players were able to understand progression, navigate the environments, and complete puzzles without requiring direct explanation. Particular attention was given to how clearly environmental elements communicated their purpose, how players reacted to asymmetric gameplay situations, and whether the intended cooperative interactions functioned as expected.

Traversal sections and puzzle sequences were tested repeatedly to identify situations where players became confused, stuck, or failed to understand the intended solution. Environmental layouts, object placement, lighting, and traversal routes were adjusted throughout development based on observations gathered during testing sessions.

Multiplayer testing was also important due to the cooperative nature of the game. Levels were tested with multiple players in order to evaluate communication flow, synchronization of gameplay events, and how well the asymmetric roles encouraged cooperation between players.

Testing was performed continuously during development rather than only at the end of implementation. This allowed gameplay issues and level flow problems to be identified early and adjusted iteratively throughout the project.

4. Conclusion

The level design of Parallax primarily focused on supporting the gameplay systems and cooperative mechanics developed throughout the project. While the levels themselves did not require large amounts of complex programming, they were essential in giving purpose and context to the implemented gameplay systems.

The environments, puzzles, and traversal sections were designed to create situations where movement, interaction, and asymmetric player abilities could be utilized in meaningful ways. Without the level design, many of the implemented gameplay systems would have lacked practical gameplay value.

A balance was therefore necessary between creating sufficiently detailed and functional gameplay environments while still maintaining a realistic development scope within the project timeframe. The focus was placed on creating levels that supported gameplay progression, cooperative interaction, and mechanic variety rather than spending excessive time on environmental complexity or visual detail alone.

Through the use of modular gameplay systems and reusable interaction mechanics, the level design was able to support multiple gameplay scenarios while remaining flexible throughout development.

Bibliography

[1] U. Technologies, "Unity Asset Store." [Online]. Available: <https://assetstore.unity.com/>